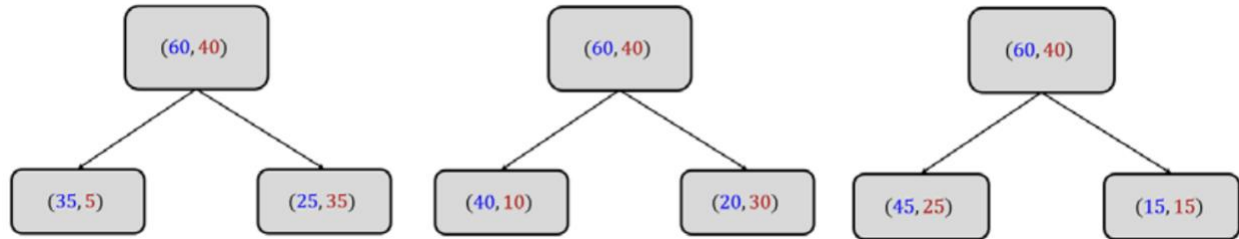


CS 767 Advanced Machine Learning Homework 5

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1. **Impurity Metrics.** Compute the change in (a) misclassification rate, (b) Gini index, and (c) Entropy for each of the following splits (full points will be given only if all intermediate steps are clearly shown).



a. Misclassification rate

Split 1:

$$L_{mc}(R_p) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{60}{100}, \frac{40}{100}\right) = 1 - \max(0.6000, 0.4000) = 1 - 0.6000 = 0.4000$$

$$L_{mc}(R_1) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{35}{40}, \frac{5}{40}\right) = 1 - \max(0.8750, 0.1250) = 1 - 0.8750 = 0.1250$$

$$L_{mc}(R_2) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{25}{60}, \frac{35}{60}\right) = 1 - \max(0.4167, 0.5833) = 1 - 0.5833 = 0.4167$$

$$\begin{aligned} \Delta L_{mc} &= 0.4000 - \left(\frac{40}{100} \cdot 0.1250 + \frac{60}{100} \cdot 0.4167\right) \\ &= 0.4000 - (0.4000 \cdot 0.1250 + 0.6000 \cdot 0.4167) \\ &= 0.4000 - (0.0500 + 0.2500) = 0.4000 - 0.3000 = 0.1000 \end{aligned}$$

Split 2:

$$L_{mc}(R_p) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{60}{100}, \frac{40}{100}\right) = 1 - \max(0.6000, 0.4000) = 1 - 0.6000 = 0.4000$$

$$L_{mc}(R_1) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{40}{50}, \frac{10}{50}\right) = 1 - \max(0.8000, 0.2000) = 1 - 0.8000 = 0.2000$$

$$L_{mc}(R_2) = 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{20}{50}, \frac{30}{50}\right) = 1 - \max(0.4000, 0.6000) = 1 - 0.6000 = 0.4000$$

$$\begin{aligned}
\Delta L_{mc} &= 0.4000 - \left(\frac{50}{100} \cdot 0.2000 + \frac{50}{100} \cdot 0.4000 \right) \\
&= 0.4000 - (0.5000 \cdot 0.2000 + 0.5000 \cdot 0.4000) \\
&= 0.4000 - (0.1000 + 0.2000) = 0.4000 - 0.3000 = 0.1000
\end{aligned}$$

Split 3:

$$\begin{aligned}
L_{mc}(R_p) &= 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{60}{100}, \frac{40}{100}\right) = 1 - \max(0.6000, 0.4000) = 1 - 0.6000 \\
&= 0.4000
\end{aligned}$$

$$\begin{aligned}
L_{mc}(R_1) &= 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{45}{70}, \frac{25}{70}\right) = 1 - \max(0.6429, 0.3571) = 1 - 0.6429 \\
&= 0.3571
\end{aligned}$$

$$\begin{aligned}
L_{mc}(R_2) &= 1 - \max(\hat{p}_c) = 1 - \max\left(\frac{15}{30}, \frac{15}{30}\right) = 1 - \max(0.5000, 0.5000) = 1 - 0.5000 \\
&= 0.5000
\end{aligned}$$

$$\begin{aligned}
\Delta L_{mc} &= 0.4000 - \left(\frac{70}{100} \cdot 0.3571 + \frac{30}{100} \cdot 0.5000 \right) \\
&= 0.4000 - (0.7000 \cdot 0.3571 + 0.3000 \cdot 0.5000) \\
&= 0.4000 - (0.2500 + 0.1500) = 0.4000 - 0.4000 = 0.0000
\end{aligned}$$

b. Gini index

Split 1:

$$\begin{aligned}
L_{gini}(R_p) &= \sum_{c=i}^c \hat{p}_c(1 - \hat{p}_c) = \frac{60}{100} \left(1 - \frac{60}{100}\right) + \frac{40}{100} \left(1 - \frac{40}{100}\right) \\
&= 0.6000(1 - 0.6000) + 0.4000(1 - 0.4000) = 0.2400 + 0.2400 = 0.4800
\end{aligned}$$

$$\begin{aligned}
L_{gini}(R_1) &= \sum_{c=i}^c \hat{p}_c(1 - \hat{p}_c) = \frac{35}{40} \left(1 - \frac{35}{40}\right) + \frac{5}{40} \left(1 - \frac{5}{40}\right) \\
&= 0.8750(1 - 0.8750) + 0.1250(1 - 0.1250) = 0.1094 + 0.1094 = 0.2188
\end{aligned}$$

$$\begin{aligned}
L_{gini}(R_2) &= \sum_{c=i}^c \hat{p}_c(1 - \hat{p}_c) = \frac{25}{60} \left(1 - \frac{25}{60}\right) + \frac{35}{60} \left(1 - \frac{35}{60}\right) \\
&= 0.4167(1 - 0.4167) + 0.5833(1 - 0.5833) = 0.2431 + 0.2431 = 0.4861
\end{aligned}$$

$$\begin{aligned}
\Delta L_{gini} &= 0.4800 - \left(\frac{40}{100} \cdot 0.2188 + \frac{60}{100} \cdot 0.4861 \right) \\
&= 0.4800 - (0.4000 \cdot 0.2188 + 0.6000 \cdot 0.4861) \\
&= 0.4800 - (0.0875 + 0.2917) = 0.4800 - 0.3792 = 0.1008
\end{aligned}$$

Split 2:

$$\begin{aligned}
L_{gini}(R_p) &= \sum_{c=i}^c \hat{p}_c(1 - \hat{p}_c) = \frac{60}{100} \left(1 - \frac{60}{100}\right) + \frac{40}{100} \left(1 - \frac{40}{100}\right) \\
&= 0.6000(1 - 0.6000) + 0.4000(1 - 0.4000) = 0.2400 + 0.2400 = 0.4800
\end{aligned}$$

$$L_{gini}(R_1) = \sum_{c=1}^c \hat{p}_c(1 - \hat{p}_c) = \frac{40}{50} \left(1 - \frac{40}{50}\right) + \frac{10}{50} \left(1 - \frac{10}{50}\right)$$

$$= 0.8000(1 - 0.8000) + 0.2000(1 - 0.2000) = 0.1600 + 0.1600 = 0.3200$$

$$L_{gini}(R_2) = \sum_{c=1}^c \hat{p}_c(1 - \hat{p}_c) = \frac{20}{50} \left(1 - \frac{20}{50}\right) + \frac{30}{50} \left(1 - \frac{30}{50}\right)$$

$$= 0.4000(1 - 0.4000) + 0.6000(1 - 0.6000) = 0.2400 + 0.2400 = 0.4800$$

$$\Delta L_{gini} = 0.4800 - \left(\frac{50}{100} \cdot 0.3200 + \frac{50}{100} \cdot 0.4800\right)$$

$$= 0.4800 - (0.5000 \cdot 0.3200 + 0.5000 \cdot 0.4800)$$

$$= 0.4800 - (0.1600 + 0.2400) = 0.4800 - 0.4000 = 0.0800$$

Split 3:

$$L_{gini}(R_p) = \sum_{c=1}^c \hat{p}_c(1 - \hat{p}_c) = \frac{60}{100} \left(1 - \frac{60}{100}\right) + \frac{40}{100} \left(1 - \frac{40}{100}\right)$$

$$= 0.6000(1 - 0.6000) + 0.4000(1 - 0.4000) = 0.2400 + 0.2400 = 0.4800$$

$$L_{gini}(R_1) = \sum_{c=1}^c \hat{p}_c(1 - \hat{p}_c) = \frac{45}{70} \left(1 - \frac{45}{70}\right) + \frac{25}{70} \left(1 - \frac{25}{70}\right)$$

$$= 0.6429(1 - 0.6429) + 0.3571(1 - 0.3571) = 0.2296 + 0.2296 = 0.4592$$

$$L_{gini}(R_2) = \sum_{c=1}^c \hat{p}_c(1 - \hat{p}_c) = \frac{15}{30} \left(1 - \frac{15}{30}\right) + \frac{15}{30} \left(1 - \frac{15}{30}\right)$$

$$= 0.5000(1 - 0.5000) + 0.5000(1 - 0.5000) = 0.2500 + 0.2500 = 0.5000$$

$$\Delta L_{gini} = 0.4800 - \left(\frac{70}{100} \cdot 0.4592 + \frac{30}{100} \cdot 0.5000\right)$$

$$= 0.4800 - (0.7000 \cdot 0.4592 + 0.3000 \cdot 0.5000)$$

$$= 0.4800 - (0.3214 + 0.1500) = 0.4800 - 0.4714 = 0.0086$$

c. Entropy

Split 1:

$$L_{ent}(R_p) = - \sum_{c=1}^c \hat{p}_c \log_2 \hat{p}_c = - \frac{60}{100} \log_2 \left(\frac{60}{100}\right) - \frac{40}{100} \log_2 \left(\frac{40}{100}\right)$$

$$= -0.6000 \log_2(0.6000) - 0.4000 \log_2(0.4000)$$

$$= -0.6000(-0.7370) - 0.4000(-1.3219) = 0.4422 + 0.5288 = 0.9710$$

$$L_{ent}(R_1) = - \sum_{c=1}^c \hat{p}_c \log_2 \hat{p}_c = - \frac{35}{40} \log_2 \left(\frac{35}{40}\right) - \frac{5}{40} \log_2 \left(\frac{5}{40}\right)$$

$$= -0.8750 \log_2(0.8750) - 0.1250 \log_2(0.1250)$$

$$= -0.8750(-0.1926) - 0.1250(-3.0000) = 0.1686 + 0.3750 = 0.5436$$

$$\begin{aligned}
L_{ent}(R_2) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{25}{60} \log_2 \left(\frac{25}{60} \right) - \frac{35}{60} \log_2 \left(\frac{35}{60} \right) \\
&= -0.4167 \log_2(0.4167) - 0.5833 \log_2(0.5833) \\
&= -0.4167(-1.2630) - 0.5833(-0.7776) = 0.5263 + 0.4536 = 0.9799 \\
\Delta L_{ent} &= 0.9710 - \left(\frac{40}{100} \cdot 0.5436 + \frac{60}{100} \cdot 0.9799 \right) \\
&= 0.9710 - (0.4000 \cdot 0.5436 + 0.6000 \cdot 0.9799) \\
&= 0.9710 - (0.2174 + 0.5879) = 0.9710 - 0.8053 = 0.1656
\end{aligned}$$

Split 2:

$$\begin{aligned}
L_{ent}(R_p) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{60}{100} \log_2 \left(\frac{60}{100} \right) - \frac{40}{100} \log_2 \left(\frac{40}{100} \right) \\
&= -0.6000 \log_2(0.6000) - 0.4000 \log_2(0.4000) \\
&= -0.6000(-0.7370) - 0.4000(-1.3219) = 0.4422 + 0.5288 = 0.9710 \\
L_{ent}(R_1) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{40}{50} \log_2 \left(\frac{40}{50} \right) - \frac{10}{50} \log_2 \left(\frac{10}{50} \right) \\
&= -0.8000 \log_2(0.8000) - 0.2000 \log_2(0.2000) \\
&= -0.8000(-0.3219) - 0.2000(-2.3219) = 0.2575 + 0.4644 = 0.7219 \\
L_{ent}(R_2) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{20}{50} \log_2 \left(\frac{20}{50} \right) - \frac{30}{50} \log_2 \left(\frac{30}{50} \right) \\
&= -0.4000 \log_2(0.4000) - 0.6000 \log_2(0.6000) \\
&= -0.4000(-1.3219) - 0.6000(-0.7370) = 0.5288 + 0.4422 = 0.9710 \\
\Delta L_{ent} &= 0.9710 - \left(\frac{50}{100} \cdot 0.7219 + \frac{50}{100} \cdot 0.9710 \right) \\
&= 0.9710 - (0.5000 \cdot 0.7219 + 0.5000 \cdot 0.9710) \\
&= 0.9710 - (0.3610 + 0.4855) = 0.9710 - 0.8464 = 0.1245
\end{aligned}$$

Split 3:

$$\begin{aligned}
L_{ent}(R_p) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{60}{100} \log_2 \left(\frac{60}{100} \right) - \frac{40}{100} \log_2 \left(\frac{40}{100} \right) \\
&= -0.6000 \log_2(0.6000) - 0.4000 \log_2(0.4000) \\
&= -0.6000(-0.7370) - 0.4000(-1.3219) = 0.4422 + 0.5288 = 0.9710 \\
L_{ent}(R_1) &= - \sum_{c=i}^c \hat{p}_c \log_2 \hat{p}_c = -\frac{45}{70} \log_2 \left(\frac{45}{70} \right) - \frac{25}{70} \log_2 \left(\frac{25}{70} \right) \\
&= -0.6429 \log_2(0.6429) - 0.3571 \log_2(0.3571) \\
&= -0.6429(-0.6374) - 0.3571(-1.4854) = 0.4098 + 0.5305 = 0.9403
\end{aligned}$$

$$\begin{aligned}
L_{ent}(R_2) &= - \sum_{c=1}^C \hat{p}_c \log_2 \hat{p}_c = -\frac{15}{30} \log_2 \left(\frac{15}{30}\right) - \frac{15}{30} \log_2 \left(\frac{15}{30}\right) \\
&= -0.5000 \log_2(0.5000) - 0.5000 \log_2(0.5000) \\
&= -0.5000(-1.0000) - 0.5000(-1.0000) = 0.5000 + 0.5000 = 1.0000 \\
\Delta L_{ent} &= 0.9710 - \left(\frac{70}{100} \cdot 0.9403 + \frac{30}{100} \cdot 1.0000 \right) \\
&= 0.9710 - (0.7000 \cdot 0.9403 + 0.3000 \cdot 1.0000) \\
&= 0.9710 - (0.6582 + 0.3000) = 0.9710 - 0.9582 = 0.0128
\end{aligned}$$

Summary of Results

Change in:	Split 1	Split 2	Split 3
Misclassification Rate	0.1000	0.1000	0.0000
Gini Index	0.1008	0.0800	0.0086
Entropy	0.1656	0.1245	0.0128

2. **Gini Index.** Suppose we have a classification problem with C classes and $\hat{p}_i$ is the proportion of Class i ($1 \leq i \leq C$) in a certain tree node.

Can the Gini index for this node be written as $L_{Gini} = 1 - \sum_{i=1}^C \hat{p}_i^2$? If yes, how does this expression follow from the one discussed in class, and if not, why not?

The expression discussed in class is:

$$L_{Gini} = \sum_{i=1}^C \hat{p}_i (1 - \hat{p}_i)$$

Distribute $\hat{p}_i$:

$$L_{Gini} = \sum_{i=1}^C (\hat{p}_i - \hat{p}_i^2)$$

Distribute Σ :

$$L_{Gini} = \sum_{i=1}^C \hat{p}_i - \sum_{i=1}^C \hat{p}_i^2$$

The sum of proportions of all classes is 1:

$$\sum_{i=1}^C \hat{p}_i = 1$$

Substitute into the previous equation:

$$L_{Gini} = 1 - \sum_{i=1}^C \hat{p}_i^2$$

3. **Variance and Standard Deviation.** Please read Example 6.8 in the textbook before answering this question. Answer questions 6.71 and 6.72.

6.71 b

- a. Not more than 100 lb

$$z = \frac{100 - 155}{20} = -2.75$$

$$P(X \leq 100) = P(Z \leq -2.75) = 0.5 - \Phi(2.75) = 0.5 - 0.4970 = 0.0030$$

$$N = 2000 \times 0.003 = 6$$

- b. Between 120 and 130 lb (inclusive)

$$z_1 = \frac{120 - 155}{20} = -1.75$$

$$z_2 = \frac{130 - 155}{20} = -1.25$$

$$P(120 \leq X \leq 130) = P(-1.75 \leq Z \leq -1.25) = \Phi(1.75) - \Phi(1.25) = 0.4599 - 0.3944 = 0.0655$$

$$N = 2000 \times 0.0655 = 131$$

- c. Between 150 and 175 lb (inclusive)

$$z_1 = \frac{150 - 155}{20} = -0.25$$

$$z_2 = \frac{175 - 155}{20} = 1$$

$$P(150 \leq X \leq 175) = P(-0.25 \leq Z \leq 1) = \Phi(0.25) + \Phi(1) = 0.0987 + 0.3413 = 0.4400$$

$$N = 2000 \times 0.44 = 880$$

- d. Greater than or equal to 200 lb

$$z = \frac{200 - 155}{20} = 2.25$$

$$P(X \geq 200) = P(Z \geq 2.25) = 0.5 - \Phi(2.25) = 0.5 - 0.4878 = 0.0122$$

$$N = 2000 \times 0.0122 = 24.4$$

6.72 Suppose the diameter d of bolts manufactured by a company is normally distributed with mean $\mu = 0.5$ cm and standard deviation $\sigma = 0.4$ cm. A bolt is considered defective if $d \leq 0.45$ cm or $d > 0.55$ cm. Find the percentage of defective bolts manufactured by the company.

$$z_1 = \frac{0.45 - 0.5}{0.4} = -0.125$$

$$z_2 = \frac{0.55 - 0.5}{0.4} = 0.125$$

$$1 - P(0.45 \leq X \leq 0.55) = 1 - P(-0.125 \leq Z \leq 0.125) = 1 - 2\Phi(0.125) = 1 - 0.0995 = 0.9005$$

4. **Covariance and Correlation Coefficient.** Answer questions 5.77 (parts (a) and (c)) and 5.91.

5.77 Consider the joint distribution of X and Y in Fig. 5-24(a).

X\Y	2	3	4	Sum
1	0.06	0.15	0.09	0.30
2	0.14	0.35	0.21	0.70
Sum	0.20	0.50	0.30	

- a. Find E (X) and E (Y).

$$E(X) = 1 \times 0.30 + 2 \times 0.70 = 1.70$$

$$E(Y) = 2 \times 0.20 + 3 \times 0.50 + 4 \times 0.30 = 3.10$$

- c. Find cov(X, Y).

$$E(XY) = 1 \times 2 \times 0.06 + 1 \times 3 \times 0.15 + 1 \times 4 \times 0.09 + 2 \times 2 \times 0.14 + 2 \times 3 \times 0.35 + 2 \times 4 \times 0.21 = 0.12 + 0.45 + 0.36 + 0.56 + 2.10 + 1.68 = 5.27$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = 5.27 - 1.70 \times 3.10 = 0$$

5.91 Let X be a random variable for which $\sigma_X \neq 0$. Show that $\rho(X, X) = 1$ and $\rho(X, -X) = -1$.

$$\rho(X, X) = \frac{\text{cov}(X, X)}{\sigma_X \sigma_X} = \frac{E\left((X - E(X))(X - E(X))\right)}{\sigma_X^2} = \frac{E\left((X - E(X))^2\right)}{\sigma_X^2} = \frac{\text{var}(X)}{\sigma_X^2} = \frac{\sigma_X^2}{\sigma_X^2} = 1$$

$$\begin{aligned} \rho(X, -X) &= \frac{\text{cov}(X, -X)}{\sigma_X \sigma_{-X}} = \frac{E\left((X - E(X))(-X - E(-X))\right)}{\sigma_X \sqrt{\text{var}(-X)}} = \frac{E\left((X - E(X))(-X + E(X))\right)}{\sigma_X \sqrt{(-1)^2 \times \text{var}(X)}} \\ &= \frac{E\left(-(X - E(X))(X - E(X))\right)}{\sigma_X \sqrt{\text{var}(X)}} = \frac{-E\left((X - E(X))^2\right)}{\sigma_X \sigma_X} = \frac{-\text{var}(X)}{\sigma_X^2} = -\frac{\sigma_X^2}{\sigma_X^2} \\ &= -1 \end{aligned}$$